What the dead code says: abandoned routes in the Lean proof of the quasi-Riemann hypothesis

Notes on the pruning of `openai/math`, `lean/OAI/NumberTheory/DirichletL`

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Abstract

The Lean development behind OpenAI's quasi-Riemann hypothesis (zero-free half-plane $\operatorname{Re} s > 7/8$ for Dirichlet and Hecke L-functions, and the shorter 23/24 proof) contains about 6,400 named declarations, roughly 99,000 source lines, that no Comparator-checked result uses. We pruned them. This note reconstructs, from the deleted code alone, what those abandoned branches were trying to do and why they were dropped. It then lists the few ideas that look worth more compute and the dead ends worth avoiding. Almost all of the deleted material is *finished* mathematics: there is no `sorry` anywhere. What it records is a search process: interfaces guessed before their inputs existed, theorems re-proved with a different quantifier order, and a handful of genuine obstructions.

1 Method and caveats

Data. Reachability was computed inside Lean, through types and proof terms, from the results that must survive:

- the 7/8 theorems for ζ, Dirichlet and Hecke L-functions;
- the new 23/24 theorems for Dirichlet L-functions and ζ;
- the one module imported from outside the directory.

A declaration counts as *dead* if it is unreachable; Table 1 gives the counts by area. Five agents each audited one area. They read the original sources at commit `fd4aeeb`, diffed dead modules against their live successors, and searched for hypotheses that nothing ever discharges. This note condenses their reports.

Caveats. The repository has no development history: it was published as a single commit. "Why it was abandoned" is therefore an inference from three kinds of evidence:

- *the diff between a dead theorem and its live twin*, usually a change of one hypothesis or one quantifier;
- *hypotheses that nothing constructs*: a structure or Prop that only ever appears as an assumption;
- *import depth*, used as a proxy for chronology.

We mark conjectures as such. "Dead" means unused by the final theorems, not wrong: every dead declaration compiles.

2 Why code died: a taxonomy

(A) Superseded copies; the majority. A module was cloned, one hypothesis or quantifier changed, and the whole chain proved again. Some examples:

Area (directories)	dead declarations	lines	files removed
Moments (7/8 fourth moments)	1,480	24,235	95
Descent, Inversion, Dictionary (7/8)	952	17,724	76
Theta function, reflection, cusps, Hecke	1,230	16,008	49
Sieves, Gauss sums, mean square	1,200	20,301	0
Detector, energy, Mellin step, extraction	1,557	20,711	33
Total	6,419	98,979	253

Table 1: Unreachable named declarations by area, and the source lines they span. Files are removed when nothing in them survives.

- In `Descent/` and `Inversion/`, 39 of 76 deleted modules have a surviving copy that shares at least 70% of its lines.
- In `Reflection/`, about 2.8k of 5.2k dead lines are twins that differ only in where an $\exists$ `degree` sits.
- In `Moments/`, the chain for the Θ-rows was rebuilt three times. The last rebuild only removed the hypothesis `IsCoprimeQC`, paying instead a constant factor $\mathrm{N}(\mathrm{rad}\, Q)^{2/3}$.

(B) Conditional chains whose hypothesis was never discharged.

- `FixedTestPrimeMS`: the prime-row extraction of the 11/12 paper. It needs many prime ideals in a norm window.
- `CanonicalLowerMoments`: a strong induction over a real parameter, which is not well-founded as stated.
- A single-cusp reflection hypothesis `hreflect`.
- Per-row ramification certificates.
- A Mertens-type prefix bound.

In every case the live proof won by *eliminating* the hypothesis, not by proving it.

(C) Unneeded strength.

- Contour estimates uniform in every parameter, where the paper says only the power margins need to be uniform.
- A full spectral theory of the Laplacian: self-adjointness, global meromorphy, a finite-dimensional theta space.
- Uniformity over a parametric family of test functions, where explicit seminorm products were needed.

(D) Genuine mathematical dead ends. These are few, and are listed in Section 5.

(E) Live by replacement. Upstream's own unconditional 23/24 assembly is dead only because the new paper-numbered top level re-proves it from the same lemmas. It was never abandoned.

3 Findings by area

3.1 The cubic theta function and its reflection

Paper 2 (the 11/12 paper) reduces the reflection at a cusp a/c to three standard cusps. It does so by asserting, without proof, that Kubota's θ is invariant under $\mathrm{SL}_2(\mathbb{Z})$. The dead code shows the formalizers meeting exactly this point.

- **The early chain could not get past the principal cusp.** It used only the Kubota group $\Gamma(3)$ (`levelThree`) and stopped there: the arithmetic structure requires $3 \mid c_0$. The Weyl-type cusps ($(c, \lambda) = 1$) need the $\mathrm{SL}_2(\mathbb{Z})$-invariance.
- **The workaround was to average.** The live proof averages θ over the 24 cosets of $\Gamma(3)$ in $\Gamma(3)\mathrm{SL}_2(\mathbb{Z})$. The average is automorphic by construction, and its Fourier coefficients are an explicit mixture $\frac{1}{4}(\theta_h + 3\theta_h^{\mathrm{opp}})$ of the expansions at ∞ and at the opposite cusp.
- **The invariance itself is never stated or proved.** No dead or live declaration asserts it, so the formal proof is correct without it.

Two further routes died:

- **The spectral and Green-identity routes to the residue.** The Green route stalled before any Maass–Selberg relation on a truncated fundamental domain. The live residue comes from a Fredholm pencil with an explicit Laurent split at $s = 4/3$; only that local interface is used.
- **An Euler-factor recurrence for $\tau(hp^3) = |p|\tau(h)$.** It solves to $A + Bq^k$, which leaves one constant free, so it cannot prove the relation. The live proof shows instead that the difference is a decaying 8/9-eigenfunction orthogonal to the residue, hence zero.

3.2 The top level of the 23/24 proof: Mellin step and extraction

- **Six interfaces for the Mellin step.** The step "power saving ⇒ zero-free half-plane" went through six successive interfaces:

  1. an exponentially smoothed rational Möbius sum;
  2. the same sum over ideals of $\mathbb{Z}[\omega]$;
  3. sharp-cutoff prefix sums;
  4. compactly supported smooth weights;
  5. the same with the Euler factors at S deleted;
  6. prime rows.

  Each is a complete reduction to a hypothesis. Each was dropped because the one analytic input ever proved, the mean-square bound, controls only smooth, compactly supported sums over ideals prime to S. The interfaces were written about 120 import levels before that input existed.
- **Paper 2's prime-row extraction was formalized but never closed.** It is conditional on `FixedTestPrimeMS`, which asks for $\gg Y^{1-\varepsilon}$ prime ideals of norm about $Y = D^{11/60}$, and nothing constructs it. The integer rows b^6, with a totient weight, replaced it. They give *exactly the same exponent* $11/12 + 5\vartheta/12$ and need no prime ideal theorem.
- **Rational-prime rows are strictly worse.** An earlier variant used rows r^6 for rational primes r. It reaches only 89/90, because $\mathrm{N}(r) = r^2$ leaves only $Y^{1/2}$ rows.
- **The prime count needed is now available.** The 7/8 chain's prime ideal count (`AnnularPrimeMass`) would discharge `FixedTestPrimeMS`, but that would add fidelity to Paper 2, not a better result.

3.3 Descent, inversion and dictionary (7/8)

There is no hidden alternative proof here: all 914 dead declarations are finished. The largest group, about 2.7k lines, is a ladder of re-proofs forced by the shape of the final induction predicate:

- windows must be chosen before the degree;

- constants must be uniform in the index types;
- the degree must be uniform in the level.

This is the 7/8 paper's lemma on finite seminorm propagation. It is stated abstractly there, and the formalization rediscovered it only when the top-level induction failed.

Other replacements:

- Path-length and real-parameter inductions became induction on an integer rank. The formal losses per step are 15η and 62η, against the paper's 11η and $40\eta+\tau+\pi$; this does not affect the exponent.
- Per-source radii and slot lists became data that is constant on each cell.
- Sums over one fixed cube became a global source, keeping the cube sum inside the square.
- Pools that depend on the tuples became the excluded-prime pool.

No variant implements Paper 2's adaptive cube cutoff H_c, and none gives a better exponent.

3.4 Fourth moments (7/8, `Moments/`)

- **The rectangle cancellation needed explicit seminorm constants.** Its constants went through four forms of uniformity before reaching explicit seminorm products. The form "$\exists J, \forall W, \exists C$" cannot feed an induction whose test functions move.
- **A decentered majorant was a dead end.** One chain majorized the rectangle difference by $|S(X)|^2 + |S(Y)|^2$. That gives up the Θ cancellation the centered range needs, although it is valid for the uncentered range $A \le 5M/6$.
- **The descent hypotheses were tuned rather than proved.** The second-transform descent's hypotheses on child energies could not be met at the recursive call. The live versions do two things:
  - restrict to children that actually occur, the "frozen" variant, at the cost of a factor $(n_2 n_3/n_0 n_1)^{1/6}$;
  - pay 1/N(seed) explicitly in the tail term, the "seeded" variant.
- **One paragraph of the 7/8 paper was formalized but is unnecessary.** The 7/8 paper's "added row zero" estimate (sixth powers among the rows) is formalized verbatim. The formal proof keeps the zero row punctured, so it never needs it.
- **There is no slack in the exponent bookkeeping.** Dead ledger lemmas certify the paper's exponent inequality, and show it is tight at $v = M/4$.

3.5 Detector, energy and the 7/8 assembly

Most of this is the "before" version of three changes in the order of choices:

- a hypothesis "for every slot datum D" became "for some fine D", as in the 7/8 paper's proposition on the order of choices;
- constants chosen inside a theorem became an input structure;
- a fixed stage loss became a parameter, together with a cap on the level.

The assembly over every zero of every character was replaced by one over primitive characters only. The uniform contour estimates were stronger than the paper requires.

3.6 Sieves, Gauss sums and the mean square

- **One conditional chain.** Version 1 of the initial dual-mean-square density, `HasInitialCanonicalDensity`, is assumed but never constructed. It demanded one derivative order for all test windows. The descent doubles the window's support radius at every level ($A \mapsto 2A + 11$), so the live version `...AtRadius` lets that order depend on the radius.
- **Two-passage transfers.** The largest cluster, about 7k lines, is a family of monolithic "first-plus-second Poisson" transfer inequalities with bespoke budget structures. They were superseded by a modular split: the first passage produces an abstract output energy, the second passage gives twisted children, and a closing step finishes. *Conjecture:* the monolithic outputs could not be fed back into the recursion without re-binning.
- **Quadratic large sieve.** No Goldmakher–Louvel attempt exists. Earlier Heath-Brown-style recursions on block norms defined by suprema stopped at a recursive inequality whose block norms were never bounded. The live proof bootstraps a single scalar exponent, $\alpha \mapsto 2 - 1/\alpha$.
- **Uniformity again.** The non-uniform versions of Prop. R and of the transfers each fail exactly one uniformity that the descent requires: uniformity in the prime pool, in the height twist, or in the window family.
- **Absent from the 7/8 directories too.** There is no band-limited weight Φ, no analogue of Paper 2's lemma removing excluded primes, and no descent with a general gap or cube cutoff. The numbers 1/20, 1/1000 and 3000 are fixed everywhere.
- **Useful leftovers.**
  - Two complete, constant-free descent lemmas: `ShortDraftDescent.integral_geometric_descent` and `binary_energy_descent`.
  - A σ-general initial invariant, `initial_invariant_power_general`.
  - The paper-faithful prime-row extraction, whose exponents are linear in ϑ.

4 Ideas worth more compute

Ranked by expected value per unit of effort.

1. **A 15/16 version of the short proof (cheap; verified prerequisite).**
   - The formal mean-square bound (`InitialMeanSquare.outside_mean_square_of_canonical_density_at_radius`) holds for every row range $D^{1+\sigma}$ with $\sigma > 1/20$, and its other input does not depend on σ. Only the extraction and the Mellin step hard-code $\vartheta = 1/10$, through the literals $11/60, 11/10, 21/10, 137/120, 23/24$.
   - Making them parametric gives a zero-free half-plane $\operatorname{Re} s > 11/12 + 5\vartheta/12$ for each $\vartheta > 1/20$, hence $\operatorname{Re} s > 15/16$ after letting ϑ tend to $1/20$.
   - The estimated cost is 100–150 Lean lines in the top-level files.
   - Reaching 11/12 itself would further require making the descent's fixed margin 1/20 a parameter (Remark 5.5 of the 23/24 paper).
2. **A parameterised descent, towards** 11/12**.**
   - Rebuild the 23/24 descent on the dead, constant-free `ShortDraftDescent.integral_geometric_descent`. Its rank should be $\lceil \log K/(c \log Z) \rceil$, with mass $C_r Z^{\varepsilon_r} (XF)^2$.
   - Seed it with `initial_invariant_power_general`, which already holds for every σ.
   - Thread the gap through the few lemmas where 1/20, 1/40, 1/1000 and 3000 enter.

Together with item 1, this is the formal route to 11/12. Its cost has not been measured.

3. **Remove the exclusion mask from the descent state (a cheap test).**

   - The live descent state carries a mask m that grows down the tree, and its invariant charges $\|m\|^2$ against the gap.
   - A Lean analogue of Paper 2's lemma removing excluded primes,

     $$E_m \le \tau(m) \sum_{d \mid m} E(K, X/\mathrm{N}d, F\mathrm{N}d),$$

     starting from the inclusion–exclusion identity `ideal_mobius_coprime_reindex`, would remove the mask.
   - That might shorten the descent. *Unverified.*

4. **Close Paper 2's $\mathrm{SL}_2(\mathbb{Z})$-invariance gap.**

   - Live and dead lemmas together reduce "θ equals its average" to one finite identity between two cubic Gauss-sum residues for horizontal h:

     $$\tfrac{9\sqrt{3}}{2} \cdot \texttt{principalArithmeticResidue}(h) = \texttt{unramifiedGaussResidue}(9h).$$

     The constant terms already agree, and both sides vanish at non-horizontal h.
   - Proving the identity would make the formal θ literally Kubota's, matching Paper 2 without averaging.
   - The "decaying eigenfunction $\Rightarrow$ 0" template already used for $\tau(hp^3)$ is an alternative route.
   - If the identity *fails*, that is itself a finding about Paper 2.

5. **Shrink the theta construction**, about 30% of the 23/24 proof's code.

   - The dead code shows that meromorphy, self-adjointness, non-vanishing of the residue and finite-dimensionality are all unused.
   - Package the Fredholm pencil and its Laurent split as one generic "simple-pole residue" lemma, then re-run the reachability audit.

6. **Shorten the 7/8 proof.**

   - Bundle the roughly 40 repeated hypotheses of the surviving descent ladder: theorem statements are 52% of its 2,191 lines.
   - Try the short, generic hybrid quadratic–cubic sieve lemmas in place of about 1,150 lines of grid machinery.
   - Use the dead decentered majorant together with `actual_live_slot_removal` for the uncentered range $A \le 5M/6$.

7. **Smaller items.**

   - Run a linear-programming slack analysis of the 7/8 exponent constraints, to find which ones bind. The known one, at $M/4$, is tight.
   - Drop the added-zero paragraph from the 7/8 paper.
   - Upstream the self-contained Mellin criteria to Mathlib.

5 Dead ends worth avoiding

Mathematics

- Reflection with $\Gamma(3)$-automorphy alone: the Weyl-type cusps need $\mathrm{SL}_2(\mathbb{Z})$. Average first, or prove the invariance first.
- Global spectral theory, or Green and Maass–Selberg identities on cusp boxes in hyperbolic 3-space: a local Laurent split at $s = 4/3$ suffices.
- Euler-factor recurrences for $\tau(hp^3)$: they leave one constant free.
- Cauchy–Schwarz over all $3^{|\varphi|}$ cusp branches: restrict to the surviving branches first.
- Large-sieve recursions on block norms defined by suprema: the scalar exponent bootstrap closes, the block-norm recursion never did.
- One monolithic two-Poisson transfer inequality with a bespoke budget: keep the passes modular.
- Decentering a rectangle difference before the Θ cancellation.
- Prime rows instead of integer rows: they give the same exponent, and need a prime ideal theorem. Rational-prime rows are strictly worse (89/90).
- Re-tuning the 7/8 exponent ledgers: they close with zero slack.

Process (for agent-driven formalization)

- **Fix the consumer before the producer.** Interfaces written before their inputs existed were rewritten six times (the Mellin step). Stage lemmas proved before the induction that consumes them were rewritten as well.
- **Settle quantifier order and uniformity first.** Most dead lines are the same theorem with $\exists$ and $\forall$ in a different order. Fix the shape of the top-level induction predicate first.
- **Check that a hypothesis can be discharged at its call site before building a chain on it.** A hypothesis-carrying structure was a warning sign every time.
- **Generalize by adding a parameter, not by copying.** Clones were usually imported by their successors, so they were also compiled on every build.
- **Prove only the uniformity the paper needs, and match its representation.** Punctured rows made one formalized paragraph of the paper unnecessary.